Maximum Likelihood Estimation (MLE)

$\mathcal{X}$: training set

θ : parameters

$$\theta^*_{MLE} = \arg\max_{\theta} p(\mathcal{X}|\theta)$$

$$p(\mathcal{X}|\theta) = \prod_{i=1}^{N} p(x_i|\theta)$$

N = 100
70 heads
30 tails
$\theta^*_{MAP} = 0.62$ (for example)

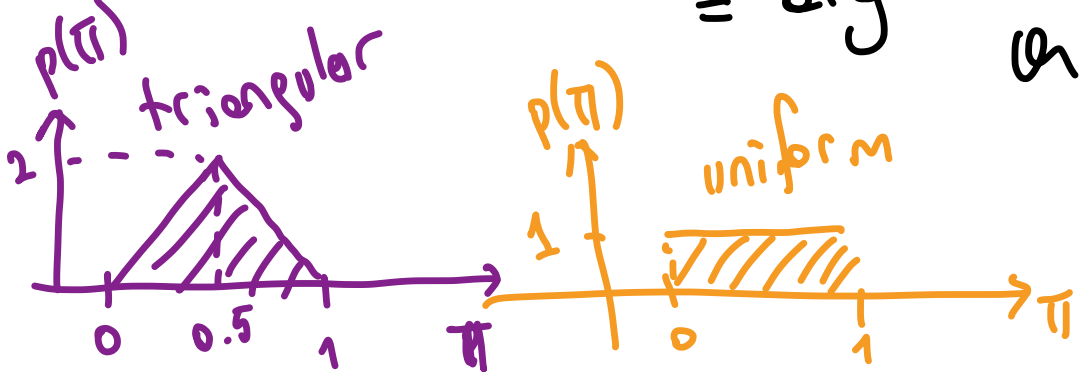
N = 1000000
700000 heads
300000 tails
$\theta^*_{MAP} = 0.69$ (for example)

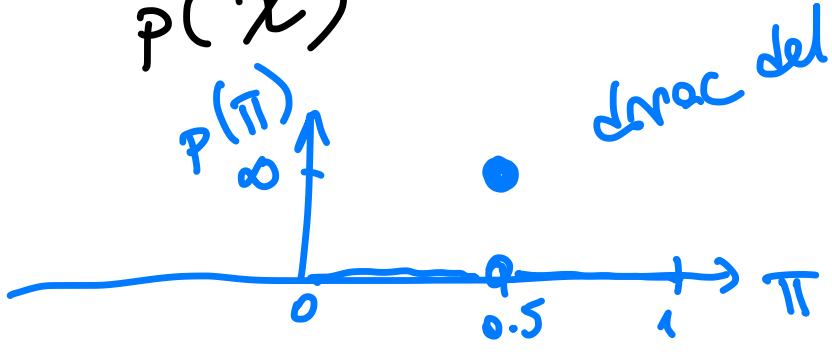
Maximum a Posteriori Estimation (MAP)

$$\theta^*_{MAP} = \arg\max_{\theta} p(\theta|\mathcal{X})$$

$$= \arg\max_{\theta} \frac{p(\mathcal{X}|\theta)\,p(\theta)}{p(\mathcal{X})}$$

→ prior belief





Perametric Regression

$$y = f(x) + \epsilon$$

y: observations; $f(x)$: underlying process; ϵ: noise

$$x_{N+1} \rightarrow \hat{y}_{N+1} = ?$$

$$f(x_{N+1}) \rightarrow \hat{y}_{N+1} \leftarrow g(x_{N+1}|\theta)$$

$\Rightarrow$ approximate $f(\cdot)$ with $g(\cdot|\theta)$

Assumptions

(I) $p(\epsilon) \sim N(\epsilon; 0, \sigma^2)$

(II) $p(y|x) \sim N(y; g(x|\theta), \sigma^2)$

$$y|x \Rightarrow f(x) + \epsilon$$

$$y|x \Rightarrow g(x|\theta) + \epsilon$$

$g(x|\theta)$: constant; ϵ: R.V.

$$E[y|x] = E[g(x|\theta) + \epsilon]$$
$$= g(x|\theta) + E[\epsilon]$$
$$= g(x|\theta)$$

$$VAR[y|x] = VAR[g(x|\theta) + \epsilon]$$
$$= 0 + VAR[\epsilon]$$
$$= \sigma^2$$

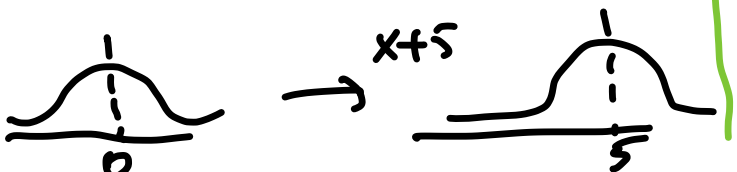
$E[X] = \delta$

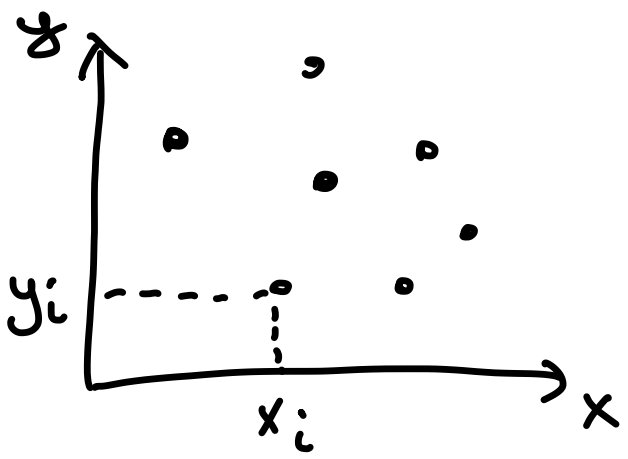
$E[X+5] = \delta + 5$

$VAR[X] = \sigma^2$

$VAR[X+5] = \sigma^2$

$x \sim N(x; 0, 9)$





$$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N} \qquad x_i \in \mathbb{R}^1 \qquad y_i \in \mathbb{R}^1$$

$$(x_i, y_i) \sim p(x_i, y_i)$$

i.i.d

$$p(x,y) = p(y|x)p(x)$$

or

$$p(x,y) = p(x|y)p(y)$$

$$p(x_1, y_1, x_2, y_2, \dots, x_N, y_N) = \prod_{i=1}^{N} p(x_i, y_i)$$

$$L(\theta|\mathcal{X}) = \prod_{i=1}^{N} \left[p(y_i|x_i)\, p(x_i) \right]$$

$$\log L(\theta|\mathcal{X}) = \log \left[\prod_{i=1}^{N} \left[p(y_i|x_i)\, p(x_i) \right] \right]$$

$$= \sum_{i=1}^{N} \left[\log[p(y_i|x_i)] + \log[p(x_i)] \right]$$

constant

maximize $\sum_{i=1}^{N} \log[p(y_i|x_i)]$ => maximize $\sum_{i=1}^{N} \log\left[N\left(y_i;\ \overbrace{g(x_i|\theta)}^{\text{mean}},\ \underbrace{\sigma^2}_{\text{variance}}\right)\right]$

R.V.

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \Rightarrow \mathcal{N}(x;\mu,\sigma^2)$$

R.V. ← x ; μ → mean ; σ² → variance

$$\text{maximize} \sum_{i=1}^{N} \log\left[\frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[-\frac{(y_i - g(x_i|\theta))^2}{2\sigma^2}\right]\right]$$

$\frac{1}{\sqrt{2\pi\sigma^2}}$ → constant

$$\text{maximize} \sum_{i=1}^{N} -\frac{(y_i - g(x_i|\theta))^2}{2\sigma^2}$$

$$\text{minimize} \sum_{i=1}^{N} (y_i - g(x_i|\theta))^2$$

$$\text{minimize} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{N} e_i^2$$

Pick a model family

$g \rightarrow$ linear

$w_0 + w_1 \cdot x_i$

$\theta = \{w_0, w_1\}$

$g \rightarrow$ quadratic

$w_0 + w_1 \cdot x_i + w_2 x_i^2$

$\theta = \{w_0, w_1, w_2\}$

Minimize $\sum_{i=1}^{N} [y_i - g(x_i|\theta)]^2$ $\quad g(x_i|\theta) = w_0 + w_1 \cdot x_i$

$$\text{Error}[\theta|\chi] = \sum_{i=1}^{N} [y_i - [w_0 + w_1 \cdot x_i]]^2$$

$$\frac{\partial \text{Error}}{\partial w_0} = \sum_{i=1}^{N} 2 \cdot [y_i - [w_0 + w_1 x_i]]^{(2-1)} \cdot (-1) = 0$$

$$\frac{\partial \text{Error}}{\partial w_1} = \sum_{i=1}^{N} 2 [y_i - [w_0 + w_1 x_i]]^{2-1} \cdot (-x_i) = 0$$

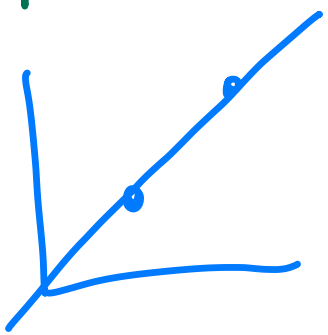
$$\sum_{i=1}^{N} y_i = \sum_{i=1}^{N} w_0 + \sum_{i=1}^{N} (w_1 x_i)$$

$$\sum_{i=1}^{N} (y_i \cdot x_i) = \sum_{i=1}^{N} (w_0 x_i) + \sum_{i=1}^{N} (w_1 x_i^2)$$

$$3x + 4y = 7$$
$$4x - 5y = 4$$

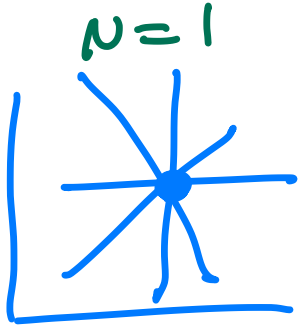
$$\begin{bmatrix} 7 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 4 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

N=2



$$\begin{bmatrix} \sum_{i=1}^{N} y_i \\ \sum_{i=1}^{N} (y_i x_i) \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{N} (1) & \sum_{i=1}^{N} x_i \\ \sum_{i=1}^{N} x_i & \sum_{i=1}^{N} x_i^2 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$$

N=1



$$A^{-1} b = A^{-1} A \quad \theta \Rightarrow \theta^{*} = A^{-1} \cdot b$$

When N=1:

$$A = \begin{bmatrix} 1 & x_1 \\ x_1 & x_1^2 \end{bmatrix} \Rightarrow \det(A) = 1 \cdot x_1^2 - x_1 \cdot x_1 = 0$$

not invertible!

When N=2:

$$A = \begin{bmatrix} 2 & (x_1+x_2) \\ (x_1+x_2) & x_1^2+x_2^2 \end{bmatrix} \Rightarrow \det(A) = 2x_1^2 + 2x_2^2 - (x_1+x_2)^2$$

$$= x_1^2 + x_2^2 - 2x_1 x_2$$

$$= (x_1 - x_2)^2 \boxed{\geq 0}$$

Exercise: Show that when $N \geq 2$, A is invertible.

Polynomial Regression:

$$g(x_i | w_0, w_1, \ldots, w_K) = w_0 + w_1 x_i + \ldots + w_K x_i^K$$

when $K=1 \Rightarrow$ linear regression

when $K=0 \Rightarrow$ constant line

$$\begin{bmatrix} N & \Sigma x_i & \cdots & \Sigma x_i^K \\ \Sigma x_i & \Sigma x_i^2 & \cdots & \Sigma x_i^{K+1} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma x_i^K & \Sigma x_i^{K+1} & \cdots & \Sigma x_i^{2K} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_K \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma(y_i x_i) \\ \Sigma(y_i x_i^2) \\ \vdots \\ \Sigma(y_i x_i^K) \end{bmatrix}$$

Can be written as $D^T D$

A: $(K+1) \times (K+1)$ — w: $(K+1) \times 1$ — b: $(K+1) \times 1$

when $K=1$

$$\begin{bmatrix} N & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma (y_i x_i) \end{bmatrix}$$

when $K=0$

$$[N][w_0] = [\Sigma y_i]$$

$$\boxed{w_0^* = \frac{\Sigma y_i}{N}} \Leftarrow \text{Sample mean}$$

Parametric Methods

Density Estimation

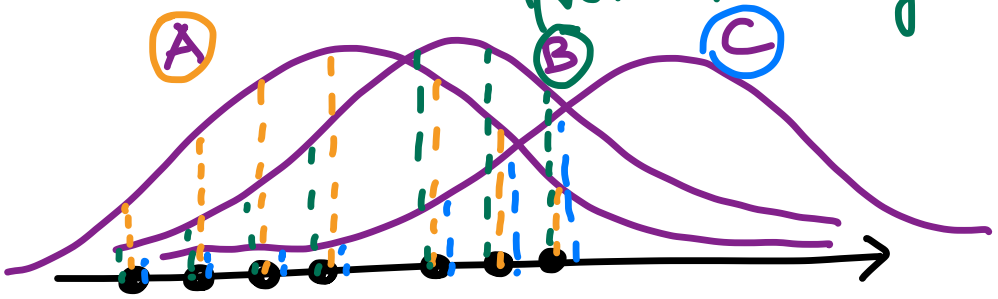
$X = \{x_i\}_{i=1}^{N}$ — N samples, N data points

$x_i \sim p(x_i) \quad \forall i \Rightarrow$ probability distribution

$\Downarrow$

parameters (?)

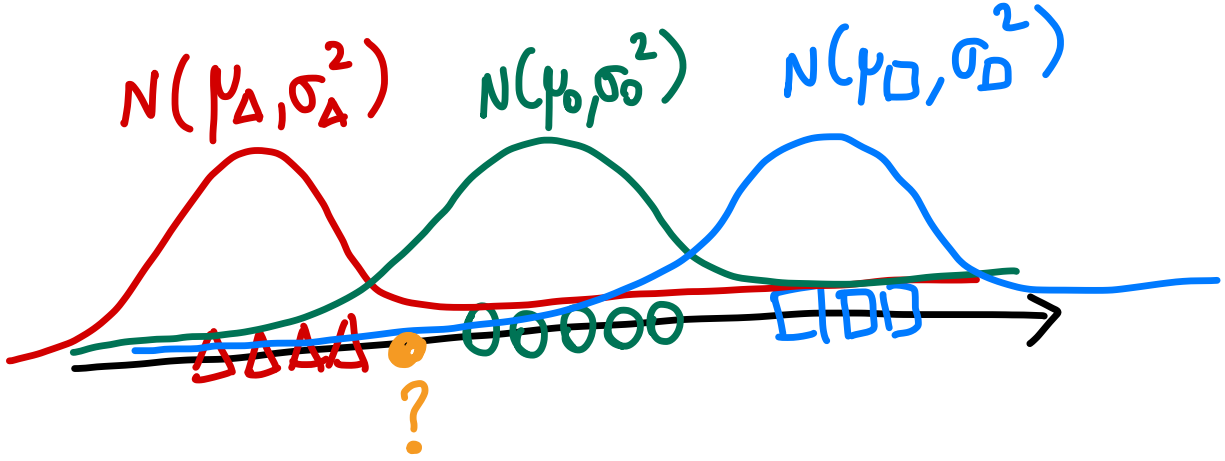
learn these parameters from training data



$x_i \sim N(x_i; \mu, \sigma^2)$

$\mu^* \Rightarrow$ best μ parameter

$\sigma^{2*} \Rightarrow$ best σ^2 parameter



$K = 3$ (# of classes)

$$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N} \qquad x_i \in \mathbb{R}^1 \qquad y_i \in \{1, 2, 3\}$$

△ ○ □

density

$p(x \mid y=c) \Rightarrow$ class conditioned density

$Pr(y=c) \Rightarrow$ prior probability

probability

$$Pr(y=c \mid x) = \frac{p(x \mid y=c)\, Pr(y=c)}{p(x)}$$

A B

posterior probability

BAYES RULE $\Rightarrow Pr(A \mid B) = \dfrac{Pr(B \mid A)\, Pr(A)}{Pr(B)}$

predicted posterior probability

$$\hat{Pr}(y=c \mid x) = \frac{\hat{p}(x \mid y=c)\, \hat{Pr}(y=c)}{\hat{p}(x)}$$

x_{N+1} $\hat{Pr}(y=c \mid x_{N+1})$

- $\hat{Pr}(y=1 \mid x_{N+1})$ (△)
- $\hat{Pr}(y=2 \mid x_{N+1})$ (○)
- $\hat{Pr}(y=3 \mid x_{N+1})$ (□)

pick the one with the maximum probability

MAXIMUM LIKELIHOOD ESTIMATION (MLE)

$\chi = \{x_i\}_{i=1}^{N} \quad x_i \sim p(x_i|\theta) \quad \forall i$

↳ unknown parameters of p

x_i's are i.i.d. (identically & independently distributed)

$Pr(A,B) = Pr(A)Pr(B)$

$P(A,B) = P(A)\,P(B)$

$\text{Likelihood} \equiv p(x_1, x_2, \ldots, x_N|\theta) \Rightarrow$ full joint

$= p(x_1|\theta)\,p(x_2|\theta)\ldots p(x_N|\theta)$ ← i.i.d.

$\mathcal{L}(\theta) \equiv \prod_{i=1}^{N} p(x_i|\theta)$

$\log\text{likelihood} = \log\left[\prod_{i=1}^{N} p(x_i|\theta)\right]$ ← natural logarithm

$\log(a.b.c) = \log(a)+\log(b)+\log(c)$

$\log(a^b) = b.\log(a)$

$= \sum_{i=1}^{N} \log\left[p(x_i|\theta)\right]$

- Gaussian → exp
- Exponential → exp
- Bernoulli → power

Bernoulli Density

$0 < \pi < 1$ ↳ success probability

(H) success : π $\Rightarrow x = 1$

(T) failure : $1-\pi$ $\Rightarrow x = 0$

coin ⇒ H T H H H H T . . . . T

$x_1\ x_2\ x_3\ x_4\ x_5\ x_6\ x_7$ . . . . x_{100}

1 0 1 1 1 1 0 0

$1-x_i$

$$p(x_i|\pi) = \pi^{x_i}(1-\pi)^{1-x_i}$$

$$Pr(x_i = 1|\pi) = \pi^1 \cdot (1-\pi)^{1-1} = \pi \quad \text{success}$$

$$Pr(x_i = 0|\pi) = \pi^0 \cdot (1-\pi)^{1-0} = 1-\pi \quad \text{failure}$$

$$\mathcal{L}(\pi|\chi) = \prod_{i=1}^{N} \left[\pi^{x_i}(1-\pi)^{(1-x_i)}\right]$$

likelihood of your training set

$$\log\text{likelihood} = \sum_{i=1}^{N} \left[x_i \log(\pi) + (1-x_i)\log(1-\pi)\right]$$

$$\frac{\partial \log\text{likelihood}}{\partial \pi} = \sum_{i=1}^{N} \left[\frac{x_i}{\pi} + \frac{(1-x_i)\cdot(-1)}{1-\pi}\right] = 0 \Rightarrow \pi^* = \frac{\sum_{i=1}^{N} x_i}{N}$$

$\frac{\#\text{of heads}}{\#\text{of tosses}}$ Sample mean

$$\frac{\partial \log(a)}{\partial a} = \frac{1}{a}$$

$$\frac{\partial \log(1-a)}{\partial a} = -\frac{1}{1-a}$$

Gaussian Density

$$\mathcal{X} = \{x_i\}_{i=1}^{N}$$

$$x_i \sim N(x_i; \mu, \sigma^2) \Rightarrow \mu^* = ? \quad , \quad \sigma^{2*} = ?$$

$$\text{log likelihood} = \log \prod_{i=1}^{N} \left[\frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[- \frac{(x_i - \mu)^2}{2\sigma^2} \right] \right]$$

$$x \sim N(x; \mu, \sigma^2)$$

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[- \frac{(x-\mu)^2}{2\sigma^2} \right]$$

$$\text{log likelihood} = \sum_{i=1}^{N} \left[-\frac{1}{2} \cdot \log(2\pi\sigma^2) - \frac{(x_i - \mu)^2}{2\sigma^2} \right]$$

$$\frac{\partial \text{log likelihood}}{\partial \mu} = 0$$

$$\frac{\partial \text{log likelihood}}{\partial \sigma^2} = 0$$

Solve for μ^* and σ^{2*}.

Solution:

$$\mu^* = \frac{\sum_{i=1}^{N} x_i}{N}$$

sample mean

$$\sigma^{2*} = \frac{\sum_{i=1}^{N} (x_i - \mu^*)^2}{N}$$

sample variance

Parametric Classification

$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$ ⇐ Input

Output ⇒ A classifier

$$\hat{y}_{N+1} = \arg\max_{c=1}^{K} g_c(x_{N+1})$$

↳ score function for class #c

$$\hat{Pr}(y=c \mid x) = \frac{\hat{p}(x \mid y=c)\,\hat{Pr}(y=c)}{\hat{p}(x)}$$

$\hat{p}(x)$ → independent of class label c

proportional to ←

$$\hat{Pr}(y=c \mid x) \;\alpha\; \hat{p}(x \mid y=c)\,\hat{Pr}(y=c)$$

equal upto a constant ←

$$\log \hat{Pr}(y=c \mid x) \overset{+}{=} \log\left[\hat{p}(x \mid y=c)\right] + \log\left[\hat{Pr}(y=c)\right]$$

$$g_c(x)$$

$-\log[\hat{p}(x)]$

$$g_c(x) = \log\left[\hat{p}(x \mid y=c)\right] + \log\left[\hat{Pr}(y=c)\right]$$

$\hat{p}(x \mid y=c) \sim N(x; \mu_c, \sigma_c^2)$; Frequency of class #c

$$= \log\left[\frac{1}{\sqrt{2\pi\sigma_c^2}} \cdot \exp\left[-\frac{(x-\mu_c)^2}{2\sigma_c^2}\right]\right] + \log\left[\hat{Pr}(y=c)\right]$$

? ? ? ?

$\mu_c^* = ?$ $\quad \sigma_c^{2*} = ?$ $\quad \hat{Pr}(y=c) = ?$

$$\mu_c^* = \frac{\left[\sum_{i=1}^{N} x_i \, 1(y_i=c)\right]}{\sum_{i=1}^{N} 1(y_i=c)}$$

$\Delta \Rightarrow 4$
$O \Rightarrow 5$
$\square \Rightarrow 3$

$$\sigma_c^{2*} = \frac{\left[\sum_{i=1}^{N} (x_i - \mu_c^*)^2 \, 1(y_i=c)\right]}{\sum_{i=1}^{N} 1(y_i=c)}$$

$$\hat{Pr}(y=c) = \frac{N_c}{N} = \frac{\sum_{i=1}^{N} 1(y_i=c)}{N}$$

one function

$$1(\cdot) = \begin{cases} 1 & \text{if TRUE} \\ 0 & \text{otherwise} \end{cases}$$